EXT 1: Functions (Ext1), F2 Polynomials (Y11)

Remainder and Factor Theorems (Ext1)
Sum, Products and Multiplicity of Roots (Ext1)

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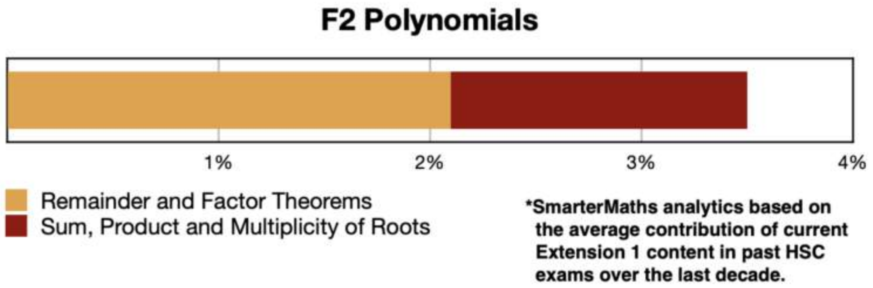
Exam Equivalent Time: 147 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*



HISTORICAL CONTRIBUTION

- *Polynomials* in the new *Ext1* course is primarily taken from the old *Ext1* course, although some content from the old *Ext2* course is also included.
- This topic has been split into two categories for analysis purposes which are: *1-Remainder and Factor Theorems (2.1%) and 2-Sum, Products and Multiplicity of Roots (1.4%).*
- This analysis looks at *Sum, Products and Multiplicity of Roots.*

HSC ANALYSIS - What to expect and common pitfalls

- *Sum, Products and Multiplicity of Roots (1.4%)* has been examined 6 times in the last 9 *Ext1* exams, primarily via multiple choice questions.
- The removal of *Polynomials* from the new *Ext2* syllabus create the possibility that *Sum, Products and Multiplicity of Roots* could be examined to a greater degree of difficulty than in the past, in our view.
- The 2020 exam tested this topic area via a multiple choice question and a very challenging cross-topic question with trigonometry (see *2020 Adv 14b*).
- Numerous *Ext2* past questions that are now examinable within the new *Ext1* course are included in the database - a number focus on the multiplicity of roots which we regard as an important revision area.

Questions

1. Functions, EXT1 F2 2016 HSC 2 MC

What is the remainder when $2x^3 - 10x^2 + 6x + 2$ is divided by $x - 2$?

(A) -66

(B) -10

(C) $-x^3 + 5x^2 - 3x - 1$

(D) $x^3 - 5x^2 + 3x + 1$

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2. Functions, EXT1 F2 2013 HSC 1 MC

The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.

What is the value of k?

(A) 2

(B) 12

(C) 20

(D) 36

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3. Functions, EXT1 F2 2017 HSC 1 MC

Which polynomial is a factor of $x^3 - 5x^2 + 11x - 10$?

A. $x - 2$

B. $x + 2$

C. $11x - 10$

D. $x^2 - 5x + 11$

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4. Functions, EXT1' F2 2015 HSC 4 MC

The polynomial $x^3 + x^2 - 5x + 3$ has a double root at $x = \alpha$.

What is the value of α?

(A) $-\dfrac{5}{3}$

(B) -1

(C) 1

(D) $\dfrac{5}{3}$

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5. Functions, EXT1 F2 2019 MET2-N 3 MC

If $x + a$ is a factor of $8x^3 - 14x^2 - a^2x$, where $a \in R, a \neq 0$, then the value of a is

A. 7

B. 4

C. 1

D. −2

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6. Functions, EXT1 F2 2015 HSC 1 MC

What is the remainder when $x^3 - 6x$ is divided by $x + 3$?

(A) -9

(B) 9

(C) $x^2 - 2x$

(D) $x^2 - 3x + 3$

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7. Functions, EXT1 F2 2012 HSC 3 MC

A polynomial equation has roots α, β, γ where

$$\alpha + \beta + \gamma = -2, \quad \alpha\beta + \alpha\gamma + \beta\gamma = 3, \quad \alpha\beta\gamma = 1.$$

Which polynomial equation has the roots α, β, and γ?

(A) $x^3 + 2x^2 + 3x + 1 = 0$

(B) $x^3 + 2x^2 + 3x - 1 = 0$

(C) $x^3 - 2x^2 + 3x + 1 = 0$

(D) $x^3 - 2x^2 + 3x - 1 = 0$

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8. Functions, EXT1' F2 2012 HSC 5 MC

The equation $2x^3 - 3x^2 - 5x - 1 = 0$ has roots α, β and γ.

What is the value of $\dfrac{1}{\alpha^3\beta^3\gamma^3}$?

(A) $\dfrac{1}{8}$

(B) $-\dfrac{1}{8}$

(C) 8

(D) -8

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9. Functions, EXT1' F2 2017 HSC 5 MC

The polynomial $p(x) = x^3 - 2x + 2$ has roots α, β and γ.

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

A. -10

B. -6

C. -2

D. 0

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10. Functions, EXT1' F2 2018 HSC 3 MC

The cubic equation $x^3 + 2x^2 + 5x - 1 = 0$ has roots, α, β and γ.

Which cubic equation has roots $\dfrac{-1}{\alpha}, \dfrac{-1}{\beta}, \dfrac{-1}{\gamma}$?

A. $x^3 - 5x^2 - 2x + 1 = 0$

B. $x^3 - 5x^2 - 2x - 1 = 0$

C. $x^3 + 5x^2 + 2x + 1 = 0$

D. $x^3 + 5x^2 - 2x + 1 = 0$

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11. Functions, EXT1' F2 2019 HSC 4 MC

The polynomial $2x^3 + bx^2 + cx + d$ has roots 1 and − 3, with one of them being a double root.

What is a possible value of b?

A. − 10

B. − 5

C. 5

D. 10

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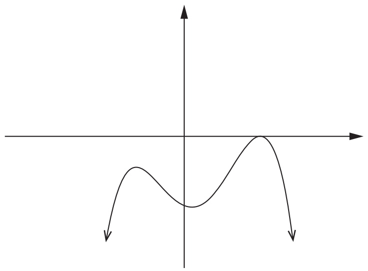
12. Functions, EXT1 F2 2020 HSC 5 MC

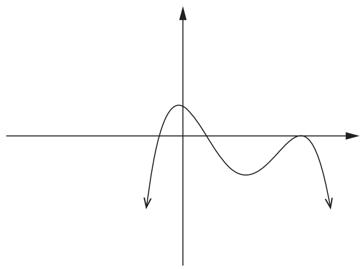
A monic polynomial $p(x)$ of degree 4 has one repeated zero of multiplicity 2 and is divisible by $x^2 + x + 1$.

Which of the following could be the graph of $p(x)$?

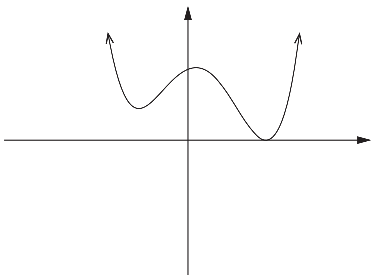
A.



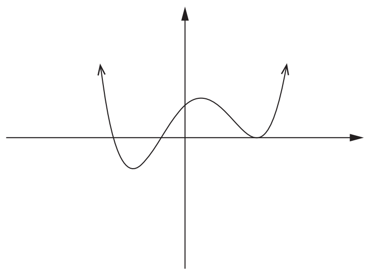
B.



C.



D.



13. Functions, EXT1 F2 2012 HSC 8 MC

When the polynomial $P(x)$ is divided by $(x+1)(x-3)$, the remainder is $2x+7$.

What is the remainder when $P(x)$ is divided by $x-3$?

(A) 1

(B) 7

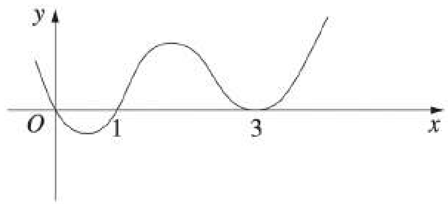
(C) 9

(D) 13

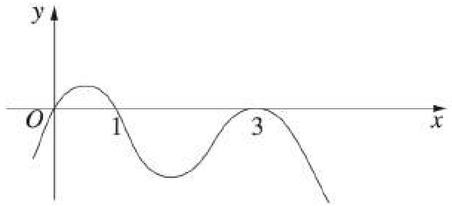
14. Functions, EXT1 F2 2013 HSC 4 MC

Which diagram best represents the graph $y = x(1-x)^3(3-x)^2$?

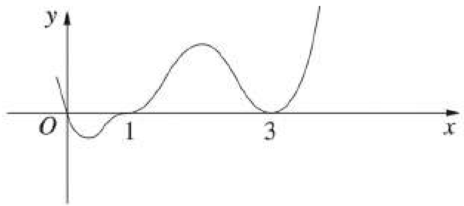
(A)



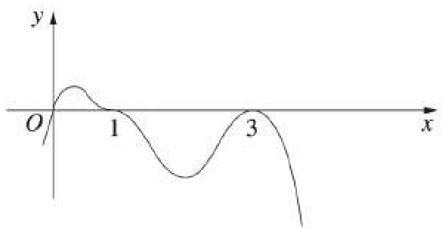
(B)



(C)



(D)



15. Functions, EXT1 F2 2014 HSC 5 MC

Which group of three numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$?

(A) $2, 3, 7$

(B) $1, -6, 7$

(C) $-1, -2, 21$

(D) $-1, -3, -14$

16. Functions, EXT1 F2 2014 HSC 9 MC

The remainder when the polynomial $P(x) = x^4 - 8x^3 - 7x^2 + 3$ is divided by $x^2 + x$ is $ax + 3$.

What is the value of a?

(A) -14

(B) -11

(C) -2

(D) 5

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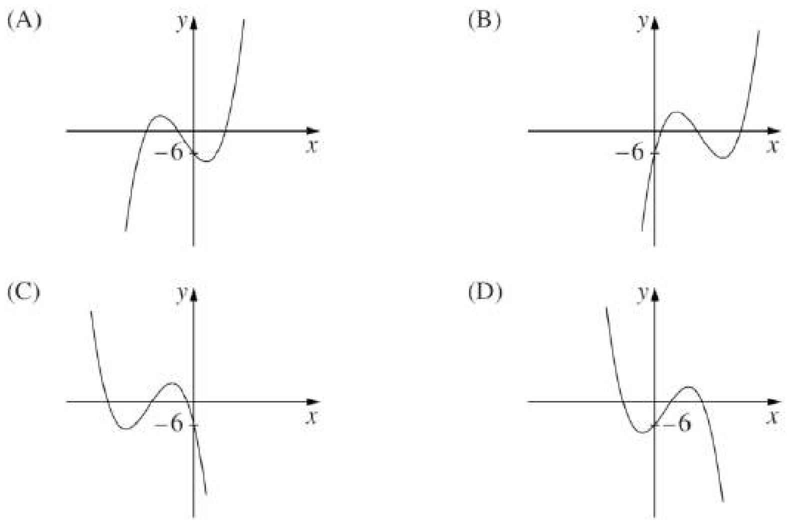
17. Functions, EXT1 F2 2016 HSC 10 MC

Consider the polynomial $p(x) = ax^3 + bx^2 + cx - 6$ with a and b positive.

Which graph could represent $p(x)$?



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18. Functions, EXT1 F2 2019 HSC 7 MC

Let $P(x) = qx^3 + rx^2 + rx + q$ where q and r are constants, $q \neq 0$. One of the zeros of $P(x)$ is -1.

Given that α is a zero of $P(x)$, $\alpha \neq -1$, which of the following is also a zero?

A. $-\dfrac{1}{\alpha}$

B. $-\dfrac{q}{\alpha}$

C. $\dfrac{1}{\alpha}$

D. $\dfrac{q}{\alpha}$

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19. Functions, EXT1' F2 2016 HSC 2 MC

Which polynomial has a multiple root at $x = 1$?

(A) $x^5 - x^4 - x^2 + 1$

(B) $x^5 - x^4 - x - 1$

(C) $x^5 - x^3 - x^2 + 1$

(D) $x^5 - x^3 - x + 1$

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20. Functions, EXT1 F2 2018 HSC 11a

Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$.

i. Show that $x = 1$ is a zero of $P(x)$. *(1 mark)*

ii. Find the other zeros. *(2 marks)*

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21. Functions, EXT1 F2 2020 HSC 11a

Let $P(x) = x^3 + 3x^2 - 13x + 6$.

i. Show that $P(2) = 0$. *(1 mark)*

ii. Hence, factor the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial. *(2 marks)*

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22. Functions, EXT1 F2 2015 HSC 11f

Consider the polynomials $P(x) = x^3 - kx^2 + 5x + 12$ and $A(x) = x - 3$.

i. Given that $P(x)$ is divisible by $A(x)$, show that $k = 6$. *(1 mark)*

ii. Find all the zeros of $P(x)$ when $k = 6$. *(2 marks)*

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23. Functions, EXT1 F2 2019 HSC 11d

Find the polynomial $Q(x)$ that satisfies $x^3+2x^2-3x-7=(x-2)Q(x)+3$. *(2 marks)*

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24. Functions, EXT1 F2 2007 HSC 2c

The polynomial $P(x)=x^2+ax+b$ has a zero at $x=2$. When $P(x)$ is divided by $x+1$, the remainder is 18.

Find the values of a and b. *(3 marks)*

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25. Functions, EXT1 F2 2009 HSC 2a

The polynomial $p(x)=x^3-ax+b$ has a remainder of 2 when divided by $(x-1)$ and a remainder of 5 when divided by $(x+2)$.

Find the values of a and b. *(3 marks)*

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26. Functions, EXT1 F2 2011 HSC 2a

Let $P(x)=x^3-ax^2+x$ be a polynomial, where a is a real number.

When $P(x)$ is divided by $x-3$ the remainder is 12.

Find the remainder when $P(x)$ is divided by $x+1$. *(3 marks)*

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27. Functions, EXT1 F2 2008 HSC 1a

The polynomial x^3 is divided by $x+3$. Calculate the remainder. *(2 marks)*

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28. Functions, EXT1 F2 2013 HSC 11a

The polynomial equation $2x^3-3x^2-11x+7=0$ has roots α, β and γ.

Find $\alpha\beta\gamma$. *(1 mark)*

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29. Functions, EXT1' F2 2014 HSC 14a

Let $P(x)=x^5-10x^2+15x-6$.

Show that $x=1$ is a root of $P(x)$ of multiplicity three. *(2 marks)*

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30. Functions, EXT1' F2 2017 HSC 13b

Let a, b and c be real numbers. Suppose that $P(x)=x^4+ax^3+bx^2+cx+1$ has roots $\alpha, \dfrac{1}{\alpha}, \beta, \dfrac{1}{\beta}$, where $\alpha>0$ and $\beta>0$.

Prove that $a=c$. *(2 marks)*

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31. Functions, EXT1' F2 2018 HSC 11b

The polynomial $p(x)=x^3+ax^2+b$ has a zero at r and a double zero at 4.

Find the values of a, b and r. *(3 marks)*

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32. Functions, EXT1 F2 2008 HSC 2c

The polynomial $p(x)$ is given by $p(x)=ax^3+16x^2+cx-120$, where a and c are constants.

The three zeros of $p(x)$ are -2, 3 and β.

Find the value of β. *(3 marks)*

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33. Functions, EXT1' F2 2017 HSC 12d

Let $P(x)$ be a polynomial.

i. Given that $(x-\alpha)^2$ is a factor of $P(x)$, show that

$$P(\alpha)=P'(\alpha)=0. \quad \textit{(2 marks)}$$

ii. Given that the polynomial $P(x)=x^4-3x^3+x^2+4$ has a factor $(x-\alpha)^2$, find the value of α. *(2 marks)*

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34. Functions, EXT1 F2 SM-Bank 2

The polynomial $P(x)=x^3-2x^2+kx+24$ has roots α, β, γ.

i. Find the value of $\alpha+\beta+\gamma$. *(1 mark)*

ii. Find the value of $\alpha\beta\gamma$. *(1 mark)*

iii. It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of k. *(2 marks)*

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35. Functions, EXT1 F2 2004 HSC 3b

Let $P(x)=(x+1)(x-3)Q(x)+a(x+1)+b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x+1)$ the remainder is -11.

When $P(x)$ is divided by $(x-3)$ the remainder is 1.

i. What is the value of b? *(1 mark)*

ii. What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$? *(2 marks)*

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36. Functions, EXT1 F2 2010 HSC 2c

Let $P(x) = (x+1)(x-3)Q(x) + ax + b$,

where $Q(x)$ is a polynomial and a and b are real numbers.

The polynomial $P(x)$ has a factor of $x-3$.

When $P(x)$ is divided by $x+1$ the remainder is 8.

i. Find the values of a and b. *(2 marks)*

ii. Find the remainder when $P(x)$ is divided by $(x+1)(x-3)$. *(1 mark)*

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37. Functions, EXT1' F2 2013 HSC 15b

The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$ has remainder -3 when divided by $x-1$. The polynomial has a double root at $x = -1$.

i. Show that $4a + 2c = -\dfrac{9}{2}$. *(2 marks)*

ii. Hence, or otherwise, find the slope of the tangent to the graph $y = P(x)$ when $x = 1$. *(1 mark)*

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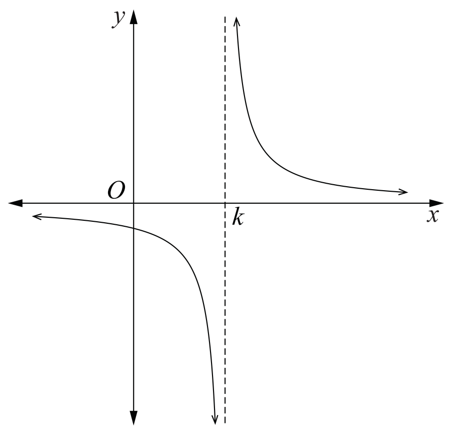
38. Functions, EXT1 F2 2019 HSC 14b

The diagram shows the graph of $y = \dfrac{1}{x-k}$, where k is a positive real number.



By considering the graphs of $y = x^2$ and $y = \dfrac{1}{x-k}$, explain why the function $f(x) = x^3 - kx^2 - 1$ has exactly one real zero. *(2 marks)*

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39. Functions, EXT1 F2 SM-Bank 1

The equation $2x^3 + x^2 - kx + 6 = 0$ has 2 roots which are reciprocals of each other.

Find the value of k. *(2 marks)*

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40. Functions, EXT1' F2 2007 HSC 3b

The zeros of $x^3 - 5x + 3$ are α, β and γ.

Find a cubic polynomial with integer coefficients whose zeros are $2\alpha, 2\beta$ and 2γ. *(2 marks)*

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41. Functions, EXT1' F2 2009 HSC 3c

Let $P(x) = x^3 + ax^2 + bx + 5$, where a and b are real numbers.

Find the values of a and b given that $(x-1)^2$ is a factor of $P(x)$. *(3 marks)*

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42. Functions, EXT1 F2 SM-Bank 5

The polynomial $P(x) = x^3 + px^2 + qx + r$ has real roots $\sqrt{k}, -\sqrt{k}$ and α.

i. Explain why $\alpha + p = 0$. *(1 mark)*

ii. Show that $k\alpha = r$. *(1 mark)*

iii. Show that $pq = r$. *(2 marks)*

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43. Functions, EXT1 F2 2006 HSC 4a

The cubic polynomial $P(x) = x^3 + rx^2 + sx + t$. where r, s and t are real numbers, has three real zeros, $1, \alpha$ and $-\alpha$.

i. Find the value of r. *(1 mark)*

ii. Find the value of $s + t$. *(2 marks)*

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44. Functions, EXT1' F2 2016 HSC 13d

Suppose $p(x) = ax^3 + bx^2 + cx + d$ with a, b, c and d real, $a \neq 0$.

i. Deduce that if $b^2 - 3ac < 0$ then $p(x)$ cuts the x-axis only once. *(2 marks)*

ii. If $b^2 - 3ac = 0$ and $p\left(-\dfrac{b}{3a}\right) = 0$, what is the multiplicity of the root $x = -\dfrac{b}{3a}$? *(2 marks)*

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45. Functions, EXT1' F2 2019 HSC 16ai

Consider the equation $x^3 - px + q = 0$, where p and q are real numbers and $p > 0$.

Let $r = \sqrt{\dfrac{4p}{3}}$ and $\cos 3\theta = \dfrac{-4q}{r^3}$.

Show that $r\cos\theta$ is a root of $x^3 - px + q = 0$.

You may use the result $4\cos^3\theta - 3\cos\theta = \cos 3\theta$. (Do NOT prove this.) *(2 marks)*

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46. Trigonometry, EXT1 T3 2020 HSC 14b

i. Show that $\sin^3\theta - \dfrac{3}{4}\sin\theta + \dfrac{\sin(3\theta)}{4} = 0$. *(2 marks)*

ii. By letting $x = 4\sin\theta$ in the cubic equation $x^3 - 12x + 8 = 0$.

Show that $\sin(3\theta) = \dfrac{1}{2}$. *(2 marks)*

iii. Prove that $\sin^2\dfrac{\pi}{18} + \sin^2\dfrac{5\pi}{18} + \sin^2\dfrac{25\pi}{18} = \dfrac{3}{2}$. *(3 marks)*

Worked Solutions

1. Functions, EXT1 F2 2016 HSC 2 MC

$$\begin{aligned} P(2) &= 2\cdot 2^3 - 10\cdot 2^2 + 6\cdot 2 + 2 \\ &= -10 \end{aligned}$$

$\Rightarrow\ B$

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2. Functions, EXT1 F2 2013 HSC 1 MC

$P(x) = x^3 - 4x^2 - 6x + k$

Since $(x-2)$ is a factor, $P(2) = 0$

$$\begin{aligned} 2^3 - 4\cdot 2^2 - 6\cdot 2 + k &= 0 \\ 8 - 16 - 12 + k &= 0 \\ k &= 20 \end{aligned}$$

$\Rightarrow\ C$

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3. Functions, EXT1 F2 2017 HSC 1 MC

$$\begin{aligned} f(2) &= 2^3 - 5\cdot 2^2 + 11\cdot 2 - 10 \\ &= 8 - 20 + 22 - 10 \\ &= 0 \end{aligned}$$

$\therefore (x-2)$ is a factor

$\Rightarrow\ A$

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4. Functions, EXT1' F2 2015 HSC 4 MC

$$\begin{aligned} P'(x) &= 3x^2 + 2x - 5 \\ &= (3x+5)(x-1) \end{aligned}$$

$\therefore$ Only possible double roots occur when

$x = 1$ or $x = -\dfrac{5}{3}$

$P(1) = 1 + 1 - 5 + 3 = 0$

$\therefore x = 1$ is double root.

$\Rightarrow\ C$

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5. Functions, EXT1 F2 2019 MET2-N 3 MC

$$\begin{aligned} f(-a) &= 8(-a)^3 - 14(-a)^2 - a^2(-a) \\ 0 &= -8a^3 - 14a^2 + a^3 \\ 0 &= -7a^3 - 14a^2 \\ 0 &= -7a^2(a+2) \\ a &= -2 \end{aligned}$$

$\Rightarrow D$

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6. Functions, EXT1 F2 2015 HSC 1 MC

$$\begin{aligned} \text{Remainder} &= P(-3) \\ &= (-3)^3 - 6(-3) \\ &= -27 + 18 \\ &= -9 \end{aligned}$$

$\Rightarrow A$

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7. Functions, EXT1 F2 2012 HSC 3 MC

Using $ax^3 + bx^2 + cx + d = 0$

By elimination

$$\alpha\beta\gamma = -\frac{d}{a} = 1$$

$\therefore$ Cannot be A or C (where $\alpha\beta\gamma = -1$)

$$\alpha + \beta + \gamma = -\frac{b}{a} = -2$$

$\therefore$ Cannot be D (where $\alpha + \beta + \gamma = 2$)

$\Rightarrow B$

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8. Functions, EXT1' F2 2012 HSC 5 MC

$$\alpha\beta\gamma = \frac{-d}{a} = \frac{1}{2}$$

$$\therefore \alpha^3\beta^3\gamma^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$\therefore \frac{1}{\alpha^3\beta^3\gamma^3} = 8$$

$\Rightarrow C$

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9. Functions, EXT1' F2 2017 HSC 5 MC

Since α, β, γ are roots,

$$\begin{aligned} p(\alpha) &= \alpha^3 - 2\alpha + 2 = 0 \quad \ldots (1) \\ p(\beta) &= \beta^3 - 2\beta + 2 = 0 \quad \ldots (2) \\ p(\gamma) &= \gamma^3 - 2\gamma + 2 = 0 \quad \ldots (3) \end{aligned}$$

Add: $(1) + (2) + (3)$

$$\alpha^3 + \beta^3 + \gamma^3 - 2(\alpha + \beta + \gamma) + 6 = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -6 \quad \left(\text{note: } \alpha + \beta + \gamma = -\frac{b}{a} = 0\right)$$

$\Rightarrow B$

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10. Functions, EXT1' F2 2018 HSC 3 MC

$\alpha\beta\gamma = 1$

$\alpha\beta + \beta\gamma + \gamma\alpha = 5$

$\alpha + \beta + \gamma = -2$

If roots are $\dfrac{-1}{\alpha}, \dfrac{-1}{\beta}, \dfrac{-1}{\gamma}$:

$$\frac{-1}{\alpha} \cdot \frac{-1}{\beta} \cdot \frac{-1}{\gamma} = \frac{-1}{\alpha\beta\gamma} = -1$$

$\Rightarrow d = 1$

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{(\alpha+\beta+\gamma)}{\alpha\beta\gamma} = -2$$

$\Rightarrow c = -2$

$$\frac{-1}{\alpha} - \frac{-1}{\beta} - \frac{-1}{\gamma} = \frac{-(\alpha\beta+\beta\gamma+\gamma\alpha)}{\alpha\beta\gamma} = -5$$

$\Rightarrow b = 5$

$\therefore$ Equation is $x^3 + 5x^2 - 2x + 1 = 0$

$\Rightarrow D$

11. Functions, EXT1' F2 2019 HSC 4 MC

$f(x) = 2x^3 + bx^2 + cx + d$

$f'(x) = 6x^2 + 2bx + c$

Roots at 1 and -3:

$f(1) = 0$

$2 + b + c + d = 0$

$b + c + d = -2 \quad \ldots (1)$

$f(-3) = 0$

$-54 + 9b - 3c + d = 0$

$9b - 3c + d = 54 \quad \ldots (2)$

$(2) - (1)$

$8b - 4c = 56$

$2b - c = 14 \quad \ldots (3)$

If double root at 1:

$f'(1) = 0$

$6 + 2b + c = 0$

$2b + c = -6 \quad \ldots (4)$

$(3) + (4)$

$4b = 8$

$b = 2$

If double root at -3:

$f'(-3) = 0$

$54 - 6b + c = 0$

$-6b + c = -54 \quad \ldots (5)$

$(3) + (5)$

$-4b = -40$

$b = 10$

$\Rightarrow D$

12. Functions, EXT1 F2 2020 HSC 5 MC

Since $p(x)$ is monic,

$\Rightarrow p(x) = (x-a)^2\left(x^2+x+1\right)$

Consider x^2+x+1

$\Delta = \sqrt{1^2 - 4\cdot1\cdot1} = \sqrt{-3} < 0 \Rightarrow$ No roots

$\therefore$ Only root is $x = a$ (multiplicity 2)

$\Rightarrow$ Eliminate B and D

As $x \to \infty,\ p(x) \to \infty$

$\Rightarrow C$

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13. Functions, EXT1 F2 2012 HSC 8 MC

Let $P(x) = A(x)\cdot Q(x) + R(x)$

where $A(x) = (x+1)(x-3)$, and $R(x) = 2x+7$

When $P(x) \div (x-3)$, remainder $= P(3)$

$$\begin{aligned} P(3) &= 0 + R(3) \\ &= (2\times3)+7 \\ &= 13 \end{aligned}$$

$\Rightarrow D$

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14. Functions, EXT1 F2 2013 HSC 4 MC

$y = x(1-x)^3(3-x)^2$

By elimination

Consider when $x < 0$,

$y = (-\text{ve})\times(+\text{ve})\times(+\text{ve}) < 0$

$\therefore$ Cannot be A or C

Consider the cubic factor $(1-x)^3$,

The graph must have a stationary point at $x = 1$

$\therefore$ Cannot be B

$\Rightarrow D$

---

15. Functions, EXT1 F2 2014 HSC 5 MC

Let roots be α, β, γ

$\alpha\beta\gamma = -\dfrac{d}{a} = -42$

$\therefore$ Cannot be A or C

$\alpha\beta + \beta\gamma + \gamma\alpha = \dfrac{c}{a} = -41$

$\therefore$ Cannot be D

$\Rightarrow B$

---

16. Functions, EXT1 F2 2014 HSC 9 MC

$P(x) = x^4 - 8x^3 - 7x^2 + 3$

$$\begin{aligned} \text{Given } P(x) &= \left(x^2+x\right)\cdot Q(x) + ax + 3 \\ &= x(x+1)Q(x) + ax + 3 \end{aligned}$$

$P(-1) = 1 + 8 - 7 + 3 = 5$

$$\begin{aligned} \therefore\ -a + 3 &= 5 \\ a &= -2 \end{aligned}$$

$\Rightarrow C$

---

17. Functions, EXT1 F2 2016 HSC 10 MC

$$p(x) = ax^3 + bx^2 + cx - 6$$
$$p'(x) = 3ax^2 + 2bx + c$$
$$p''(x) = 6ax + 2b$$

Since $a, b > 0$,
As $x \to \infty,\ \ p(x) \to \infty$
$\therefore$ Eliminate C and D.

POI occurs when $p''(x) = 0$,

$6ax + 2b = 0$

$$x = -\frac{b}{3a} < 0, \quad (a, b > 0)$$

$\therefore$ Eliminate B
$\Rightarrow\ A$

18. Functions, EXT1 F2 2019 HSC 7 MC

Roots: $\alpha,\ \beta,\ -1$

$$\alpha\beta(-1) = -\frac{d}{a} = -\frac{q}{q} = -1$$

Mean mark 51%.

$$-\alpha\beta = -1$$

$$\therefore \beta = \frac{1}{\alpha}$$

$\Rightarrow\ C$

19. Functions, EXT1' F2 2016 HSC 2 MC

Consider C,

$$P(1) = 1 - 1 - 1 + 1 = 0$$
$$P'(x) = 5x^4 - 3x^2 - 2x$$
$$P'(1) = 5 - 3 - 2 = 0$$

$\therefore$ Multiple root at $x = 1$
$\Rightarrow C$

20. Functions, EXT1 F2 2018 HSC 11a

i. $P(1) = 1 - 2 - 5 + 6 = 0$

$\therefore x = 1$ is a zero

ii. Using part (i) $\Rightarrow (x-1)$ is a factor of $P(x)$

$$P(x) = (x-1) \cdot Q(x)$$

By long division:

$$\begin{aligned} P(x) &= (x-1)\left(x^2 - x - 6\right) \\ &= (x-1)(x-3)(x+2) \end{aligned}$$

$\therefore$ Other zeroes are:
$x = -2$ and $x = 3$

21. Functions, EXT1 F2 2020 HSC 11a

i. $$\begin{aligned} P(2) &= 8 + 12 - 26 + 6 \\ &= 0 \end{aligned}$$

ii.

$$\begin{array}{r|l} & x^2 + 5x - 3 \\ \hline x - 2 & x^3 + 3x^2 - 13x + 6 \\ & x^3 - 2x^2 \\ \hline & 5x^2 - 13x + 6 \\ & 5x^2 - 10x \\ \hline & -3x + 6 \\ & -3x + 6 \\ \hline & 0 \end{array}$$

$$\therefore P(x) = (x-2)\left(x^2 + 5x - 3\right)$$

22. Functions, EXT1 F2 2015 HSC 11f

i. $P(x) = x^3 - kx^2 + 5x + 12$

$A(x) = x - 3$

If $P(x)$ is divisible by $A(x)$,

$P(3) = 0$

$$3^3 - k(3^2) + 5 \times 3 + 12 = 0$$
$$27 - 9k + 15 + 12 = 0$$
$$9k = 54$$
$$\therefore k = 6 \ \ldots \text{as required}$$

ii. Find all roots of $P(x)$

$$\begin{aligned}\therefore P(x) &= x^3 - 6x^2 + 5x + 12\\ &= (x-3)(x^2 - 3x - 4)\\ &= (x-3)(x-4)(x+1)\end{aligned}$$

$\therefore$ Zeros at $x = 3, 4, -1$

---

23. Functions, EXT1 F2 2019 HSC 11d

$$(x-2)\cdot Q(x) + 3 = x^3 + 2x^2 - 3x - 7$$
$$(x-2)\cdot Q(x) = x^3 + 2x^2 - 3x - 10$$

$$\begin{array}{r|l} & x^2 + 4x + 5 \\ \hline x-2 & x^3 + 2x^2 - 3x - 10 \\ & x^3 - 2x^2 \\ \hline & 4x^2 - 3x - 10 \\ & 4x^2 - 8x \\ \hline & 5x - 10 \\ & 5x - 10 \\ \hline & 0 \end{array}$$

$\therefore Q(x) = x^2 + 4x + 5$

---

24. Functions, EXT1 F2 2007 HSC 2c

$P(x) = x^2 + ax + b$

Since there is a zero at $x = 2$,

$$P(2) = 0$$
$$2^2 + 2a + b = 0$$
$$2a + b = -4 \quad \ldots (1)$$

$$P(-1) = 18,$$
$$(-1)^2 - a + b = 18$$
$$-a + b = 17 \quad \ldots (2)$$

Subtract $(1) - (2)$,

$$3a = -21$$
$$a = -7$$

Substitute $a = -7$ into (1),

$$2(-7) + b = -4$$
$$b = 10$$

$\therefore a = -7$ and $b = 10$

---

25. Functions, EXT1 F2 2009 HSC 2a

$$\begin{aligned} p(x) &= x^3 - ax + b \\ P(1) &= 2 \\ 1 - a + b &= 2 \\ b &= a + 1 \quad ...\ (1) \\ P(-2) &= 5 \\ -8 + 2a + b &= 5 \\ 2a + b &= 13 \quad ...\ (2) \end{aligned}$$

Substitute $b = a + 1$ into (2)

$$\begin{aligned} 2a + a + 1 &= 13 \\ 3a &= 12 \\ \therefore a &= 4 \\ \therefore b &= 5 \end{aligned}$$

26. Functions, EXT1 F2 2011 HSC 2a

$P(x) = x^3 - ax^2 + x$

Since $P(x) \div (x - 3)$ has remainder 12,

$$\begin{aligned} P(3) = 3^3 - a \times 3^2 + 3 &= 12 \\ 27 - 9a + 3 &= 12 \\ 9a &= 18 \\ a &= 2 \end{aligned}$$

$\therefore\ P(x) = x^3 - 2x^2 + x$

Remainder $P(x) \div (x + 1)$ is $P(-1)$

$$\begin{aligned} P(-1) &= (-1)^3 - 2(-1)^2 - 1 \\ &= -4 \end{aligned}$$

27. Functions, EXT1 F2 2008 HSC 1a

Using remainder theorem,

$$\begin{aligned} P(-3) &= (-3)^3 \\ &= -27 \end{aligned}$$

MARKER'S COMMENT: "Grave concern" that many who found $P(-3) = -27$ stated the remainder was 27.

$\therefore$ Remainder when $x^3 \div (x + 3) = -27$

28. Functions, EXT1 F2 2013 HSC 11a

$P(x) = 2x^3 - 3x^2 - 11x + 7 = 0$

$\alpha\beta\gamma = -\dfrac{d}{a} = -\dfrac{7}{2}$

29. Functions, EXT1' F2 2014 HSC 14a

$P(x) = x^5 - 10x^2 + 15x - 6$

$P(1) = 1 - 10 + 15 - 6 = 0$

$P'(x) = 5x^4 - 20x + 15$

$P'(1) = 5 - 20 + 15 = 0$

$P''(x) = 20x^3 - 20$

$P''(1) = 20 - 20 = 0$

$P'''(x) = 60x^2$

$P'''(1) = 60 \neq 0$

$\therefore x = 1$ is a root of $P(x)$, of multiplicity 3.

30. Functions, EXT1' F2 2017 HSC 13b

Sum of roots $= -\dfrac{b}{a}$:

$$-a = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$$

Sum of products of 3 roots $= -\dfrac{d}{a}$:

$$\begin{aligned}-c &= \alpha \cdot \frac{1}{\alpha} \cdot \beta + \alpha \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} + \alpha \cdot \beta \cdot \frac{1}{\beta} + \frac{1}{\alpha} \cdot \beta \cdot \frac{1}{\beta} \\ &= \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha} \\ &= -a\end{aligned}$$

$\therefore a = c \ldots$ as required.

31. Functions, EXT1' F2 2018 HSC 11b

$p(x) = x^3 + ax^2 + b$

$p'(x) = 3x^2 + 2ax$

Double root at 4:

$$\begin{aligned}p'(4) &= 0 \\ 3 \times 16 + 8a &= 0 \\ a &= -6\end{aligned}$$

$$\begin{aligned}p(4) &= 0 \\ 64 + 16a + b &= 0 \\ 64 - 96 + b &= 0 \\ b &= 32\end{aligned}$$

Roots of $p(x)$ are $4, 4, r$

$$4 + 4 + r = -\frac{a}{1} = 6$$

$$\therefore r = 2$$

32. Functions, EXT1 F2 2008 HSC 2c

$p(x) = ax^3 + 16x^2 + cx - 120$

Roots: $-2,\ 3,\ \beta$

$$\begin{aligned}-2 + 3 + \beta &= -\frac{B}{A} \\ \beta + 1 &= -\frac{16}{a} \\ \beta &= -\frac{16}{a} - 1 \quad \text{... (1)}\end{aligned}$$

$$\begin{aligned}-2 \times 3 \times \beta &= -\frac{D}{A} \\ -6\beta &= \frac{120}{a} \\ \beta &= -\frac{20}{a} \quad \text{... (2)}\end{aligned}$$

$$\begin{aligned}-\frac{16}{a} - 1 &= -\frac{20}{\beta} \\ -16 - a &= -20 \\ a &= 4\end{aligned}$$

MARKER'S COMMENT: Many students displayed significant inefficiencies in solving simultaneous equations.

Substitute $a = 4$ into (1)

$$\begin{aligned}\therefore \beta &= -\frac{16}{4} - 1 \\ &= -5\end{aligned}$$

33. Functions, EXT1' F2 2017 HSC 12d

i. $P(x) = (x-\alpha)^2 \cdot Q(x)$

$$P'(x) = 2(x-\alpha) \cdot Q(x) + (x-\alpha)^2 \cdot Q'(x)$$
$$= (x-\alpha)[2Q(x) + (x-\alpha) \cdot Q'(x)]$$

$$P(\alpha) = 0 \times Q(x) = 0$$
$$P'(\alpha) = 0[2Q(x) + 0 \times Q'(x)] = 0$$

$\therefore P(\alpha) = P'(\alpha) = 0 \ldots$ as required.

ii. $P(x) = x^4 - 3x^3 + x^2 + 4$

$$P'(x) = 4x^3 - 9x^2 + 2x$$
$$= x\left(4x^2 - 9x + 2\right)$$
$$= x(4x-1)(x-2)$$

$\therefore P'(x) = 0$ when $x = 0, \dfrac{1}{4}$ or 2

$\Rightarrow$ Multiple roots may exist at $x = 0, \dfrac{1}{4}$ or $2.$

Test each root in $P(x)$:

$$P(0) = 0 - 0 + 0 + 4 = 4$$
$$P\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)^4 - 3\left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^2 + 4 = 4\frac{5}{256}$$
$$P(2) = 16 - 3(8) + 4 + 4 = 0$$

$\therefore (x-2)^2$ is a factor of $P(x)$

$\therefore \alpha = 2$

34. Functions, EXT1 F2 SM-Bank 2

i. $\alpha + \beta + \gamma = -\dfrac{b}{a} = 2$

ii. $\alpha\beta\gamma = -\dfrac{d}{a} = -24$

iii. Let roots be $\alpha, -\alpha, \beta$:

$$\alpha - \alpha + \beta = 2$$
$$\beta = 2$$

Substitute $\beta = 2$ into equation:

$$2^3 - 2 \cdot 2^2 + 2k + 24 = 0$$
$$2k = -24$$
$$k = -12$$

35. Functions, EXT1 F2 2004 HSC 3b

i. $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$

$$P(-1) = -11$$
$$-11 = (-1+1)(-1-3)Q(x) + a(-1+1) + b$$
$$-11 = (0)(-4)Q(x) + a(0) + b$$
$$\therefore b = -11$$

(ii) $P(3) = 1$

$$\therefore (3+1)(3-3)Q(x) + a(3+1) - 11 = 1$$
$$4a = 12$$
$$a = 3$$

When $P(x)$ is divided by $(x+1)(x-3)$

$$R(x) = a(x+1) + b$$
$$= 3(x+1) - 11$$
$$= 3x + 3 - 11$$
$$= 3x - 8$$

36. Functions, EXT1 F2 2010 HSC 2c

i. $P(x) = (x+1)(x-3)Q(x) + ax + b$

$(x-3)$ is a factor (given)

$\therefore P(3) = 0$

$3a + b = 0 \quad \ldots (1)$

$P(x) \div (x+1) = 8$ (given)

$\therefore P(-1) = 8$

$-a + b = 8 \quad \ldots (2)$

Subtract $(1)-(2)$

$$4a = -8$$
$$a = -2$$

Substitute $a = -2$ into (1)

$$-6 + b = 0$$
$$b = 6$$

$\therefore a = -2,\ b = 6$

ii. $P(x) \div (x+1)(x-3)$

$$= \frac{(x+1)(x-3)Q(x) - 2x + 6}{(x+1)(x-3)}$$

$$= Q(x) + \frac{-2x+6}{(x+1)(x-3)}$$

$\therefore$ Remainder is $-2x + 6$

COMMENT: This question requires a fundamental understanding of the remainder theorem. Ensure you are comfortable with this solution.

---

37. Functions, EXT1' F2 2013 HSC 15b

i. $P(x) = ax^4 + bx^3 + cx^2 + e$

$P'(x) = 4ax^3 + 3bx^2 + 2cx$

$P(1) = -3$

$a + b + c + e = -3 \quad \ldots (1)$

$P(-1) = 0$

$a - b + c + e = 0 \quad \ldots (2)$

$P'(-1) = 0$

$-4a + 3b - 2c = 0 \quad \ldots (3)$

$(1) - (2)$

$$2b = -3$$
$$b = -\frac{3}{2}$$

Substitute into (3)

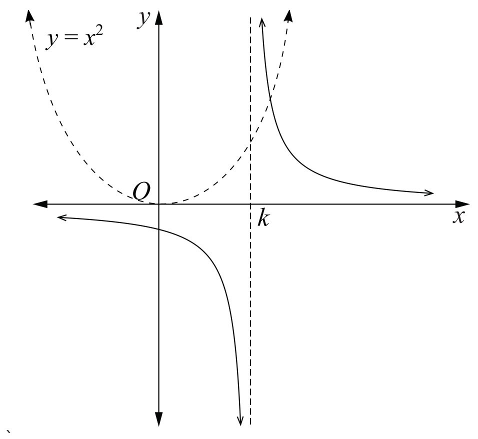
$$\therefore 4a + 2c = 3b$$
$$= -\frac{9}{2} \quad \ldots \text{ as required}$$

ii. $P'(1) = 4a + 3b + 2c$

$$= -\frac{9}{2} - \frac{9}{2}$$
$$= -9$$

38. Functions, EXT1 F2 2019 HSC 14b

Draw $y = x^2$ on the diagram:



One point of intersection occurs when

$$x^2 = \frac{1}{x-k}$$

$$x^3 - kx^2 = 1$$

$$x^3 - kx^2 - 1 = 0$$

Since only 1 point of intersection

$\Rightarrow x^3 - kx^2 - 1 = 0$ has exactly 1 zero

---

39. Functions, EXT1 F2 SM-Bank 1

Let roots be $\alpha,\ \dfrac{1}{\alpha},\ \beta$:

$$\alpha \cdot \frac{1}{\alpha} \cdot \beta = -\frac{d}{a}$$

$$\beta = -\frac{6}{2}$$

$$= -3$$

COMMENT: Substituting $\beta = -3$ into the equation solves this question efficiently.

Substitute $\beta = -3$ into equation:

$$2(-3)^3 + (-3)^2 - (-3)k + 6 = 0$$

$$-54 + 9 + 3k + 6 = 0$$

$$3k = 39$$

$$k = 13$$

---

40. Functions, EXT1' F2 2007 HSC 3b

Solution 1

$$\begin{aligned} \alpha+\beta+\gamma &= 0 \\ \therefore 2\alpha+2\beta+2\gamma &= 2(\alpha+\beta+\gamma) \\ &= 0 \\ \alpha\beta+\beta\gamma+\gamma\alpha &= -5 \\ \therefore 2\alpha 2\beta+2\beta 2\gamma+2\gamma 2\alpha &= 4(\alpha\beta+\beta\gamma+\gamma\alpha) \\ &= -20 \\ \alpha\beta\gamma &= -3 \\ \therefore 2\alpha 2\beta 2\gamma &= 8\alpha\beta\gamma \\ &= -24 \end{aligned}$$

$\therefore$ Polynomial is $x^3-20x+24$

Solution 2

$P(x)=x^3-5x+3$

New zeros are $2\alpha, 2\beta$ and 2γ

$$\begin{aligned} \therefore H(x) &= \left(\frac{x}{2}\right)^3-5\left(\frac{x}{2}\right)+3 \\ &= \frac{x^3}{8}-\frac{5x}{2}+3 \end{aligned}$$

$\therefore$ New Polynomial with integer coefficients is

$x^3-20x+24$

41. Functions, EXT1' F2 2009 HSC 3c

$$\begin{aligned} P(x) &= x^3+ax^2+bx+5 \\ P(1) &= 1+a+b+5=0 \\ \therefore b &= -a-6 \quad \text{... (1)} \end{aligned}$$

$$\begin{aligned} P'(x) &= 3x^2+2ax+b \\ P'(1) &= 3+2a+b=0 \\ 2a+b &= -3 \quad \text{... (2)} \end{aligned}$$

Substitute $b=-a-6$ from (1) into (2)

$2a+(-a-6)=-3$

$\therefore a=3, \quad b=-9$

42. Functions, EXT1 F2 SM-Bank 5

i. Sum of roots:

$$\begin{aligned} \sqrt{k}-\sqrt{k}+\alpha &= -\frac{b}{a} \\ \alpha &= -p \\ \alpha+p &= 0 \end{aligned}$$

ii. Product of roots:

$$\begin{aligned} \sqrt{k}\times -\sqrt{k}\times\alpha &= -\frac{d}{a} \\ -k\alpha &= -r \\ \therefore k\alpha &= r \end{aligned}$$

iii. $$\begin{aligned} \sqrt{k}(-\sqrt{k})+\sqrt{k}\alpha-\sqrt{k}\alpha &= \frac{c}{d} \\ -k &= q \end{aligned}$$

Substitute $k=-q$ into part (ii):

$$\begin{aligned} -q\alpha &= r \\ -q\times -p &= r \quad \text{(using part(i))} \\ \therefore pq &= r \end{aligned}$$

43. Functions, EXT1 F2 2006 HSC 4a

i. $P(x)=x^3+rx^2+sx+t$

Roots are $1, \alpha, -\alpha$

$$1+\alpha+-\alpha=-\frac{b}{a}$$
$$1=-\frac{r}{1}$$
$$\therefore r=-1$$

ii. $P(x)=x^3-x^2+sx+t$

$P(1)=0$

$0=1-1+(s\times 1)+t$

$0=s+t$

$\therefore$ Value of $s+t=0$

44. Functions, EXT1' F2 2016 HSC 13d

i. $p(x)=ax^3+bx^2+cx+d$

$p'(x)=3ax^2+2bx+c$

$\Rightarrow p(x)$ will cut the x-axis once only if $\Delta(p'(x))<0$

$$(2b)^2-4(3a)c<0$$
$$4b^2-12ac<0$$
$$b^2-3ac<0$$

ii. $p\left(-\dfrac{b}{3a}\right)=0$

$$p'\left(-\frac{b}{3a}\right)=3a\left(-\frac{b}{3a}\right)^2+2b\left(-\frac{b}{3a}\right)+c$$
$$=-\frac{b^2}{3a}+c=0\quad(\text{given } b^2-3ac=0)$$

$\therefore$ Multiplicity at least 2.

$p''(x)=6ax+2b$

$$p''\left(-\frac{b}{3a}\right)=6a\left(-\frac{b}{3a}\right)+2b=0$$

$\therefore$ Multiplicity of $x=-\dfrac{b}{3a}$ is 3.

♦♦ Mean mark 32%.

45. Functions, EXT1' F2 2019 HSC 16ai

$x^3-px+q=0\ldots(1)$

$$r=\sqrt{\frac{4p}{3}}\Rightarrow p=\frac{3r^2}{4}$$

♦ Mean mark 49%.

$$\cos 3\theta=\frac{-4q}{r^3}\Rightarrow q=\frac{-r^3\cos 3\theta}{4}$$

Given $4\cos^3\theta-3\cos\theta=\cos 3\theta\ldots(2)$

Substitute $x=r\cos\theta$ into (1)

$$r^3\cos^3\theta-\frac{3r^2}{4}r\cos\theta-\frac{r^3\cos 3\theta}{4}=0$$
$$\frac{r^3}{4}\underbrace{\left(4\cos^3\theta-3\cos\theta-\cos 3\theta\right)}_{=0\text{ (see (2) above)}}=0$$

$\therefore r\cos\theta$ is a root.

46. Trigonometry, EXT1 T3 2020 HSC 14b

i. Prove: $\sin^3\theta - \frac{3}{4}\sin\theta + \frac{\sin(3\theta)}{4} = 0$

$$\begin{aligned}
\text{LHS} &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(\sin 2\theta\cos\theta + \cos 2\theta\sin\theta) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta\cos^2\theta + \sin\theta\left(1 - 2\sin^2\theta\right)\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta\left(1 - \sin^2\theta\right) + \sin\theta - 2\sin^3\theta\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}\left(2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta\right) \\
&= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{3}{4}\sin\theta - \sin^3\theta \\
&= 0
\end{aligned}$$

ii. Show $\sin(3\theta) = \frac{1}{2}$

Using part (i):

$$\frac{\sin(3\theta)}{4} = \frac{3}{4}\sin\theta - \sin^3\theta$$

$$\sin(3\theta) = 3\sin\theta - 4\sin^3\theta \ldots (1)$$

$x^3 - 12x + 8 = 0$

Let $x = 4\sin\theta$

$$\begin{aligned}
(4\sin\theta)^3 - 12(4\sin\theta) + 8 &= 0 \\
64\sin^3\theta - 48\sin\theta &= 0 \\
-16\underbrace{\left(3\sin\theta - 4\sin^2\theta\right)}_{\text{see (1) above}} &= -8 \\
-16\sin(3\theta) &= -8 \\
\sin(3\theta) &= \frac{1}{2}
\end{aligned}$$

♦♦♦ *Mean mark part (iii) 21%.*

iii. Prove: $\sin^2\frac{\pi}{18} + \sin^2\frac{5\pi}{18} + \sin^2\frac{25\pi}{18} = \frac{3}{2}$

Solutions to $x^3 - 12x + 8 = 0$ are

$x = 4\sin\theta$ where $\sin(3\theta) = \frac{1}{2}$

When $\sin 3\theta = \frac{1}{2}$,

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \ldots$$

$$\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \ldots$$

$\therefore$ Solutions

$$x = 4\sin\frac{\pi}{18} \quad \left(= 4\sin\frac{17\pi}{18}\right)$$

$$x = 4\sin\frac{5\pi}{18} \quad \left(= 4\sin\frac{13\pi}{18}\right)$$

$$x = 4\sin\frac{25\pi}{18} \quad \left(= 4\sin\frac{29\pi}{18}\right)$$

If roots of $x^3 - 12x + 8 = 0$ are α, β, γ:

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -12$$

$$\left(4\sin\frac{\pi}{18}\right)^2 + \left(4\sin\frac{5\pi}{18}\right)^2 + \left(4\sin\frac{25\pi}{18}\right)^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$16\left(\sin^2\frac{\pi}{18} + \sin^2\frac{5\pi}{18} + \sin^2\frac{25\pi}{18}\right) = 0 - 2(-12)$$

$$\therefore \frac{\sin^2\pi}{18} + \sin^2\frac{5\pi}{18} + \frac{\sin^2(25\pi)}{18} = \frac{24}{16} = \frac{3}{2}$$